

# Functional Analysis of Omics Data

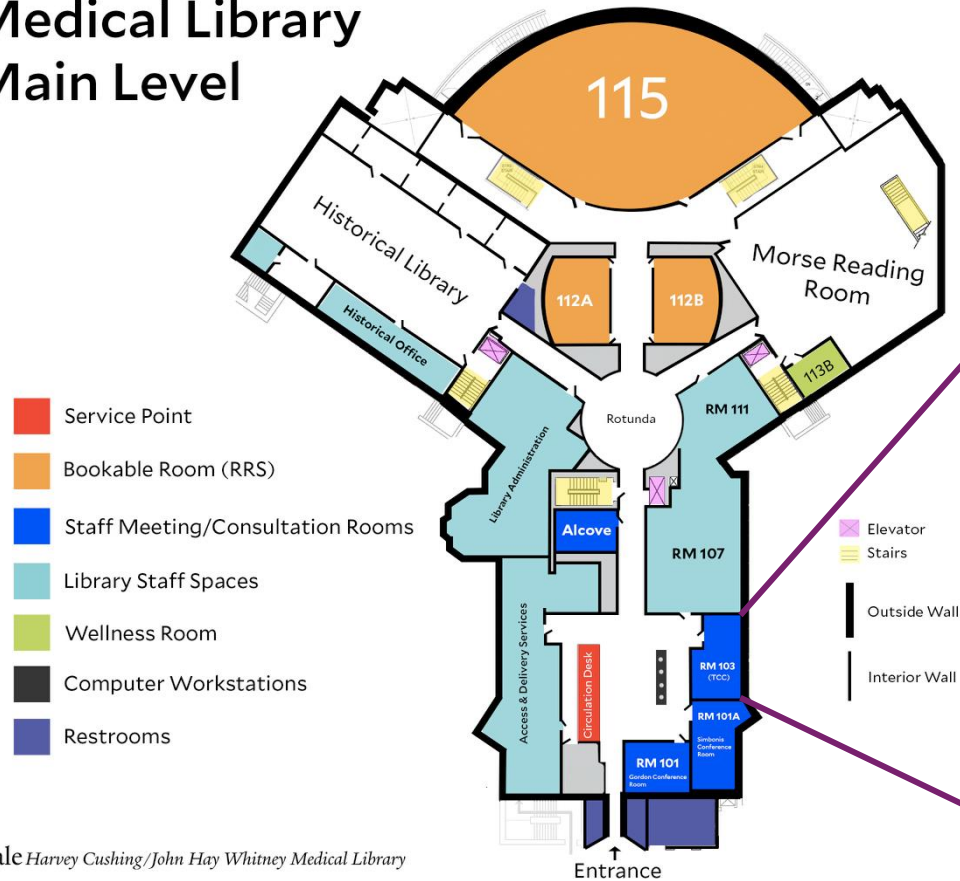
## Protein Interaction Network Analysis and Visualization

Rolando Garcia-Milian, MLS, AHIP  
Bioinformatics Support Program  
Rolando.milian@yale.edu



# Bioinformatics Support Hub (BSH) supporting research since 2014

## Medical Library Main Level



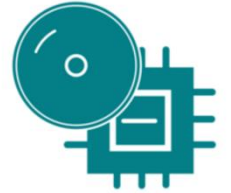
Yale Harvey Cushing/John Hay Whitney Medical Library



Consultations & Collaborations



Training



Software & Hardware

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### About

The Bioinformatics Support Program provides free access to state-of-the-art commercial bioinformatics software programs and databases for the mining, analysis, functional annotation, and visualization of Next Generation Sequencing data such as Partek Flow, QluCore Omics Explorer, Ingenuity Pathway Analysis, Ingenuity Variant Analysis MetaCore/MetaDrug, Clarivate Drug Intelligence, Biocyc, and TRANSFAC.

It also provides free individual and group instruction and consultation on how to use these software programs as well as many other public databases, data repositories and online tools such as those of the National Center for Biotechnology Information, European Bioinformatics Institute, The Cancer Genome Atlas, and others.

### Who We Are



**Rolando Garcia-Milian, MLS, AHIP**

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# Instruction and Training



## Workshop series



## Hands-on



## Demos

September 10 – October 6, 2026

## Bioinformatics Newsletter



## Yale Connect



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>SEP 8</b> Tue, 10:00am - 11:30am <b>In-Person</b> <b>WAITLIST OPEN</b><br/>Medical School, SHM L 111, Cushing/Whitney Medical School</p> <p><b>Code with Clarity: AI-Powered Programming for Researchers</b></p> <p>In this training session, we will introduce Yale's Clarity AI system and other AI-powered coding tools designed to enhance your programming workflow. Through live demonstrations and examples, you'll... <a href="#">More</a></p> <p>...</p>                                 | <p><b>SEP 8</b> Tue, 12:00pm - 12:30pm <b>Online</b> <b>20 SEATS LEFT</b><br/>Medical School</p> <p><b>Finding Research Data at Yale</b></p> <p>Yale provides many rich data resources, so how do you find out if there is data somewhere that answers your research question? This interactive session introduces the biomedical and health sciences... <a href="#">More</a></p> <p>■ Data</p>                                                                                                               | <p><b>SEP 23</b> Wed, 10:00am - 12:00pm <b>In-Person</b> <b>6 SEATS LEFT</b><br/>SHM L 111, Cushing/Whitney Medical School</p> <p><a href="#">Show more dates &gt;&gt;</a></p> <p><b>Mastering Differential Gene Expression Analysis with DESeq2 in Positron: From Demo to Your Data...</b></p> <p>Registration required Class 1 (you must attend Class 1 in order to take Class 2)* and In-Person Summary: This two-part workshop is designed to equip you with the essential skills for performing... <a href="#">More</a></p> <p>■ Bioinformatics</p> | <p><b>SEP 24</b> Thu, 10:00am - 12:00pm <b>In-Person</b><br/>SHM L 111, Cushing/Whitney Medical School</p> <p><a href="#">Show more dates &gt;&gt;</a></p> <p><b>Mastering Differential Gene Expression Analysis with DESeq2 in Positron: From Demo to Your Data...</b></p> <p>Registration required for Class 1 (you must attend Class 1 in order to take Class 2)* and In-Person Summary: This two-part workshop is designed to equip you with the essential skills for performing... <a href="#">More</a></p> <p>■ Bioinformatics</p> |
| <p><b>SEP 10</b> Thu, 10:00am - 12:00pm <b>In-Person</b> <b>11 SEATS LEFT</b><br/>Medical School, SHM L 111, Cushing/Whitney Medical School</p> <p><b>Network Visualization to Interpret Pathway Analysis Results with Enrichment Map</b></p> <p>Pathway (gene set) enrichment analyses often produce long lists of overlapping and redundant gene sets, which can make results difficult to interpret and act on. Enrichment Map addresses this by... <a href="#">More</a></p> <p>■ Bioinformatics</p> | <p><b>SEP 16</b> Wed, 10:00am - 12:00pm <b>In-Person</b> <b>14 SEATS LEFT</b><br/>Medical School, SHM L 111, Cushing/Whitney Medical School</p> <p><b>From Differential Expression to Mechanism: Protein-Protein Interaction Network Analysis...</b></p> <p>Differential expression analysis (e.g., DESeq2, limma) is powerful for identifying molecules whose expression changes across conditions. However, it often yields long gene lists that can be... <a href="#">More</a></p> <p>■ Bioinformatics</p> | <p><b>SEP 29</b> Tue, 9:00am - 9:30am <b>In-Person</b><br/>Medical School, Zoom Session -Medical (online)</p> <p><b>Webinar: QluCore Omics Explorer.Import, preprocess and analyze Olink® data</b></p> <p>Register here In this webinar presented by QluCore, you will learn how to go through the steps required to import and analyze Olink® data, from smaller to large protein panels. We will look at the... <a href="#">More</a></p> <p>■ Bioinformatics</p>                                                                                       | <p><b>OCT 6</b> Tue, 10:00am - 11:00am <b>In-Person</b><br/>Medical School, Zoom Session -Medical (online)</p> <p><b>Webinar: From Transcription Factors, Master Regulators and Disease Biomarkers to Omics Data...</b></p> <p>TRANSFAC® (Transcription Factor Database) is a comprehensive, manually curated database detailing eukaryotic transcription factors, their specific genomic binding sites, and DNA-binding profiles... <a href="#">More</a></p> <p>■ Bioinformatics</p>                                                    |
| <p><b>SEP 17</b> Thu, 10:00am - 12:00pm <b>In-Person</b> <b>14 SEATS LEFT</b><br/>Medical School, SHM L 111, Cushing/Whitney Medical School</p> <p><b>The Very Basics of the Unix Command Line</b></p> <p>This 2-hour session will introduce you to: - Navigating a *nix operating system with a terminal. - Basics of running a command. - Essential command line tools for reading and editing files. ... <a href="#">More</a></p> <p>...</p>                                                         | <p><b>SEP 22</b> Tue, 10:00am - 12:00pm <b>In-Person</b> <b>14 SEATS LEFT</b><br/>Medical School, SHM L Cushing Center</p> <p><b>Bring Your Data: Hands-On Introduction to Data Preparation and Coding</b></p> <p>Have a dataset but not sure where to start? Or no dataset yet? Either way, we'd love to have you! Join us for a hands-on, beginner-friendly coding session led by library staff. In this interactive... <a href="#">More</a></p> <p>■ Data</p>                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

# Instruction and Training



## Peer-to-peer teaching

Not professional teachers or instructors but member of the community

Cognitive congruence: awareness of current challenge of learners

MEDICAL REFERENCE SERVICES QUARTERLY  
2022, VOL. 41, NO. 1, 13–25  
<https://doi.org/10.1080/02763869.2022.2020568>



### Peer Teaching as Bioinformatics Training Strategy: Incentives, Challenges, and Benefits

Nur-Taz Rahman<sup>a</sup> , Caitlin Meyer<sup>a</sup> , Durga Thakral<sup>b</sup>, Wesley L. Cai<sup>c</sup> ,  
Ann T. Chen<sup>d</sup>, Razib Obaid<sup>e</sup> , and Rolando Garcia-Milian<sup>a</sup> 

<sup>a</sup>Bioinformatics Support Program, Research and Education Services, Harvey Cushing/John Hay Whitney Medical Library, Yale University, New Haven, Connecticut, USA; <sup>b</sup>Department of Dermatology, Yale School of Medicine, New Haven, Connecticut, USA; <sup>c</sup>Department of Pathology, Yale School of Medicine, New Haven, Connecticut, USA; <sup>d</sup>Department of Biomedical Engineering, Yale University, New Haven, Connecticut, USA; <sup>e</sup>RARAF Radiological Research Accelerator Facility, Nevis Laboratory, Columbia University, Irvington, New York, USA

## Main entrance of Yale School of Medicine



“Those having torches will pass them on to one another.”

# Instruction and Training



## Collaborations

10-sessions workshop series  
30/ 67 preselected attendees



## March 2026 Intro to NGS data analysis workshop **schedule** Every Tuesday from 9-12

Yale Connect



- Day 1. **Introduction to Bioinformatics Tools and Jupyter Notebooks** (Tue Mar 24)
- Day 2. **Project Management, Reproducibility, and Introduction to R with the Tidyverse** (Tue Mar 31)
- Day 3. **Experimental Design and Quality Control** (Tue Apr 7)
- Day 4. **Bulk RNA-seq Analysis (Single-Factor Design)** (Tue Apr 14)
- Day 5. **Bulk RNA-seq Analysis (Multi-Factor Design)** (Tue Apr 21)
- Day 6. **Pathway and Functional Enrichment Analysis** (Tue Apr 28)
- Day 7. **Single-Cell RNA-seq Analysis with Seurat (Single Sample)** (Tue May 5)
- Day 9. **Single-Cell RNA-seq Analysis with Seurat (Multiple Samples)** (Tue May 12)
- Day 10. **Somatic Variant Calling** (Tue May 26)

# Instruction and Training



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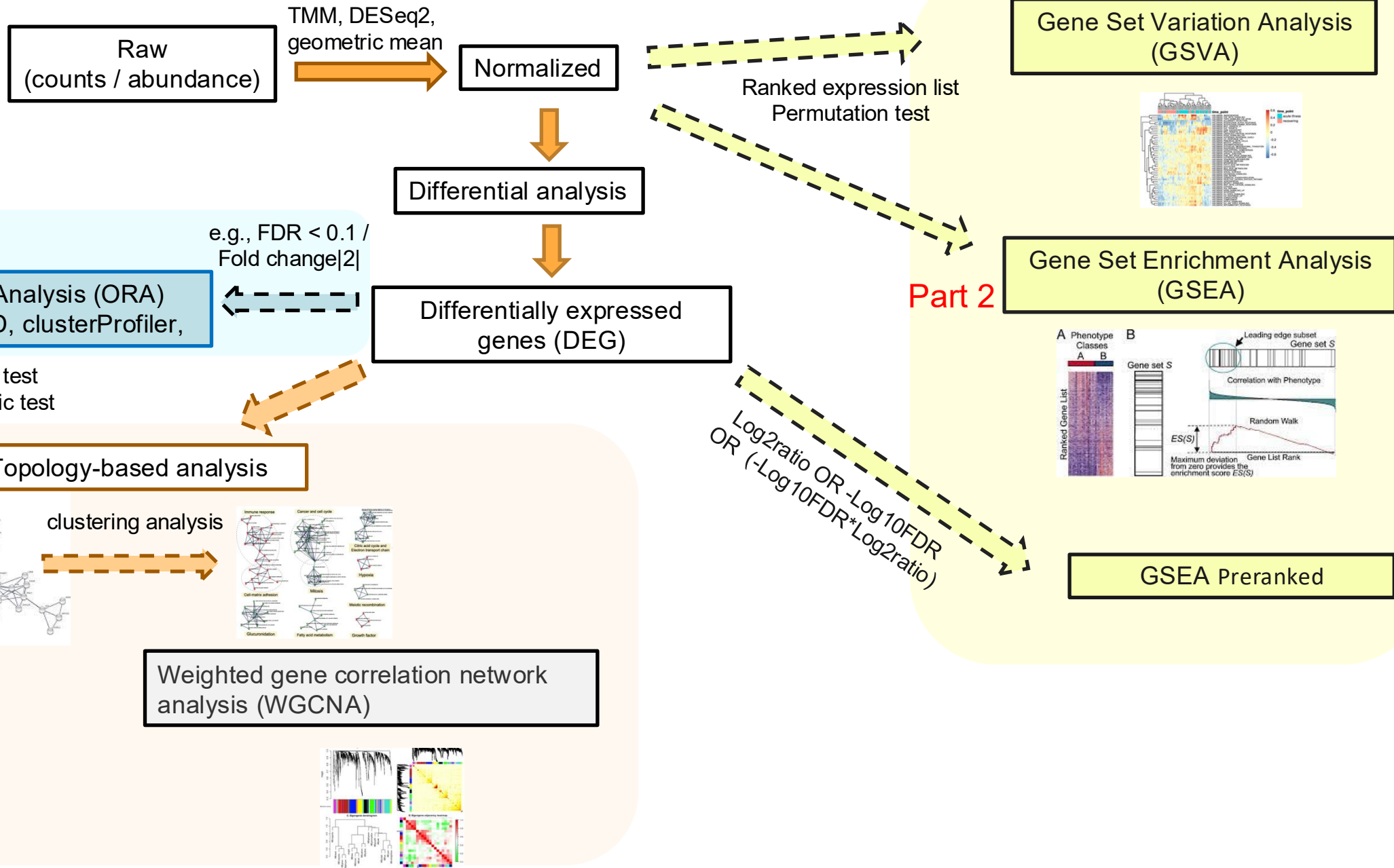


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# Session Goals

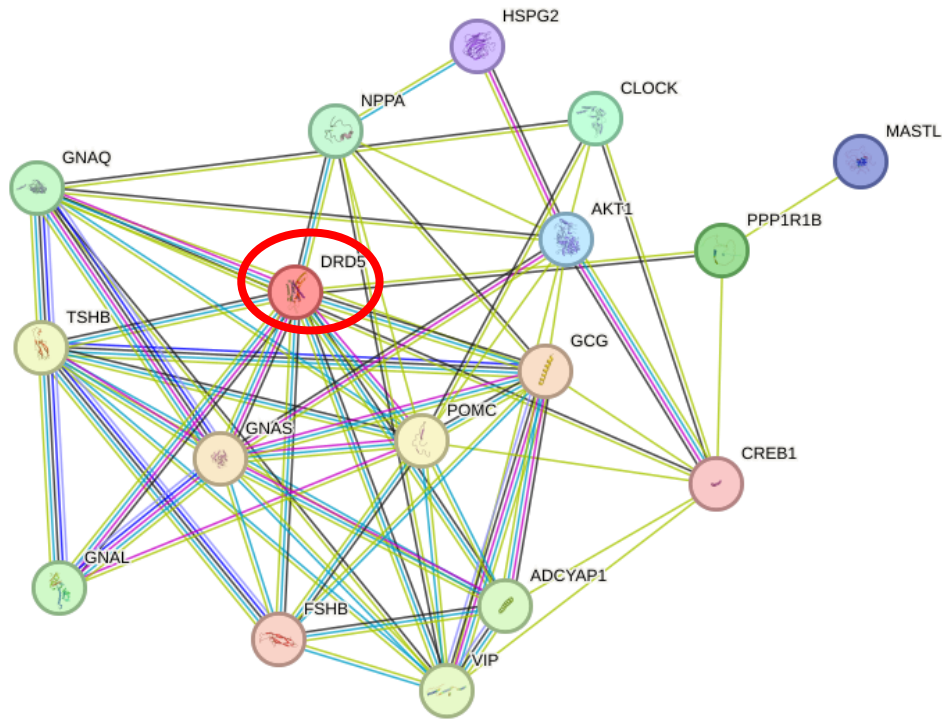
- Loading differential expression result to Cytoscape with the stringApp
- Protein-interaction (PPI) network enrichment analysis with stringApp, Enrichment Map, and Autoannotate
- PPI network clustering with stringApp and functional annotation
- Finding top hubs of network (cytoHubba app)
- Overlaying expression values on the network

# Types of enrichment analysis



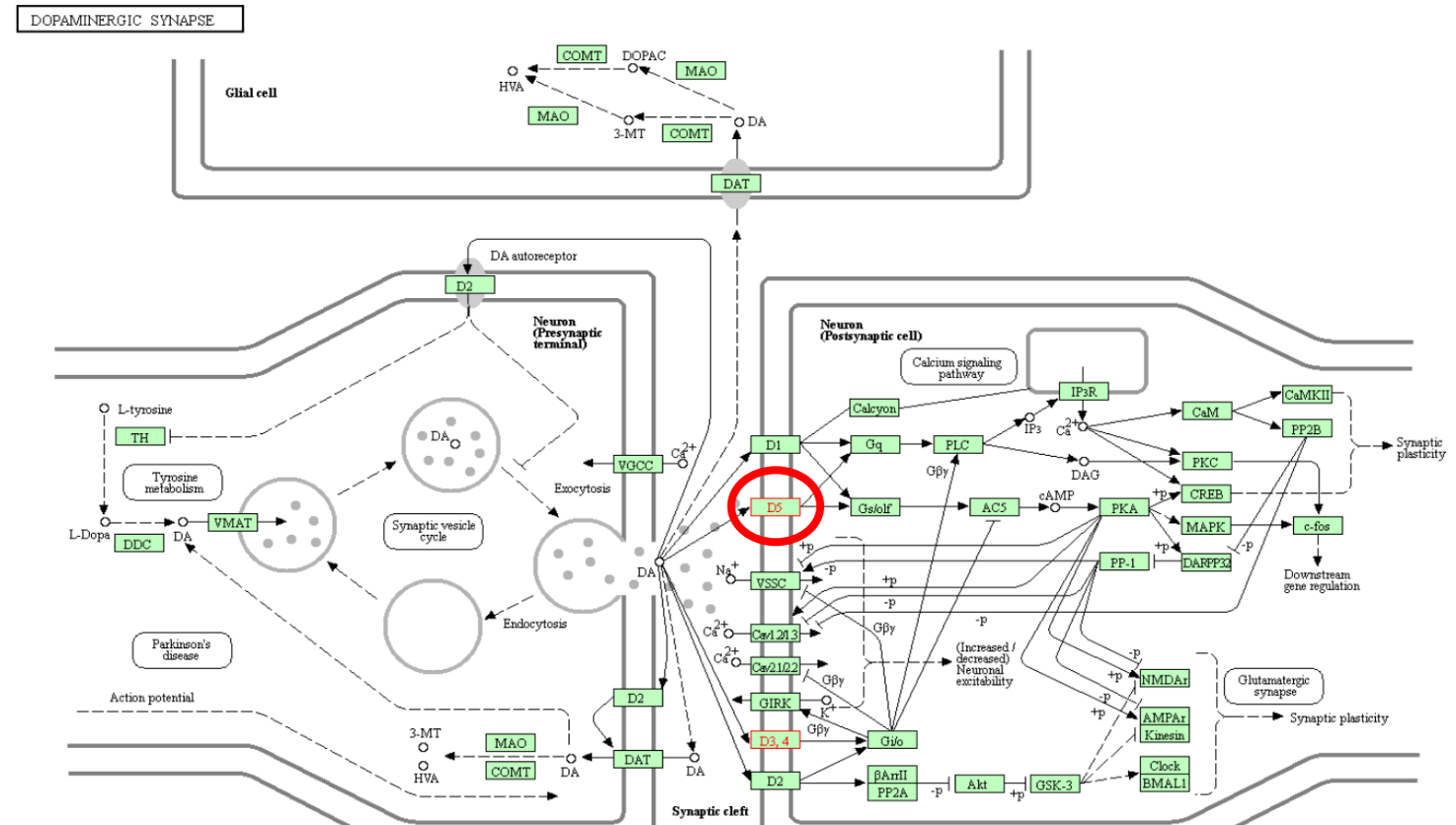
# Pathway or network?

## Dopamine receptor interaction network



STRING database

## Dopaminergic synapse pathway



Kyoto Encyclopedia of Genes and Genomes (KEGG)

# GO Annotations for human dopamine receptor

## Filter results

Total annotations: **66**

### User filters

– evidence\_subset\_closure\_label: physical interaction evidence used in manual assertion

### Your search is pinned to these filters

– document\_category: annotation  
– bioentity: UniProtKB:P21918

### Ontology (aspect)

### Organism

### Type

### Evidence

- ☒ (35) evidence used in automatic assertion
- ☒ (13) direct assay evidence used in manual assertion
- ☒ (13) experimental evidence
- ☒ (8) author statement supported by traceable reference used in manual assertion
- ☒ (6) biological aspect of ancestor evidence used in manual assertion
- ☒ (6) similarity evidence used in manual assertion
- ☒ (4) author statement without traceable support used in manual assertion

### GO class (including "regulates")

### GO class (excluding "regulates")

### GO class (direct)

### Annotation qualifier

Total annotations: 66; showing: 1-66

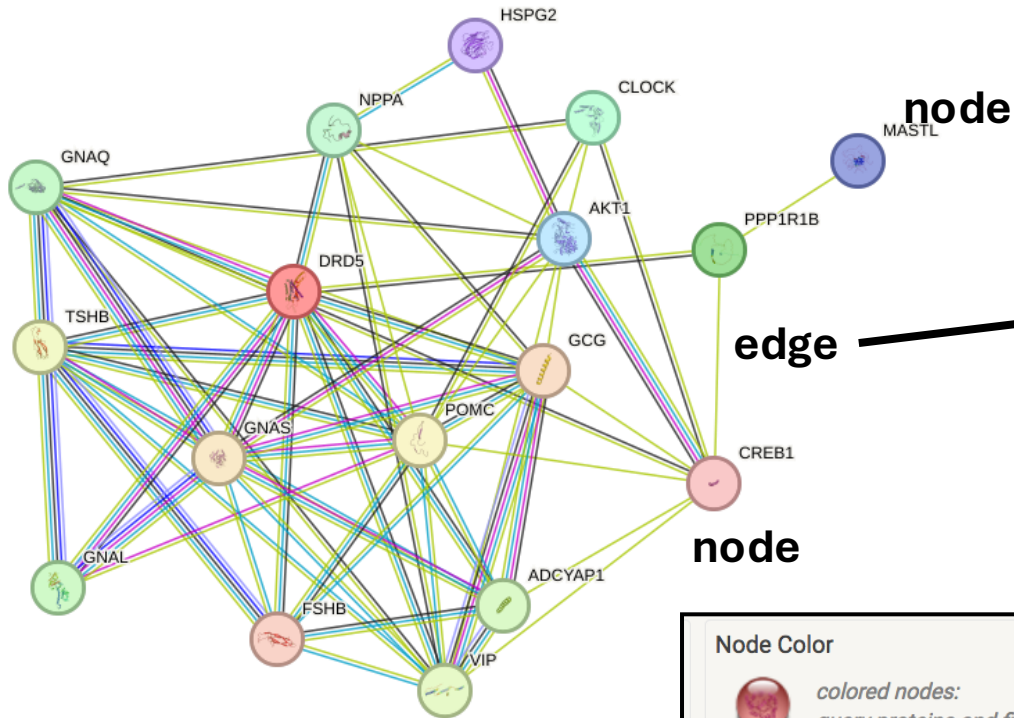
Results count **100**

«First <Prev Next> Last» Download (up to 100000)

| <input type="checkbox"/> Gene/product | name                    | qualifier | GO class (direct)                                                                                      | Annotation extension | Contributor | Organism     | Evidence | Evidence with                                                    | PANTHER family                                                   | Type    | Isoform          | Reference      |
|---------------------------------------|-------------------------|-----------|--------------------------------------------------------------------------------------------------------|----------------------|-------------|--------------|----------|------------------------------------------------------------------|------------------------------------------------------------------|---------|------------------|----------------|
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | positive regulation of MAPK cascade                                                                    |                      | GO_Central  | Homo sapiens | IBA      | PANTHER:PTN001205212<br>UniProtKB:P13945 <a href="#">more...</a> | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein |                  | GO_REF:0000033 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | dopamine neurotransmitter receptor activity, coupled via Gs                                            |                      | GO_Central  | Homo sapiens | IBA      | PANTHER:PTN008610765<br>MGI:MGI:94927                            | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein |                  | GO_REF:0000033 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | G protein-coupled receptor activity                                                                    |                      | GO_Central  | Homo sapiens | IBA      | PANTHER:PTN002798190<br>MGI:MGI:106673 <a href="#">more...</a>   | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein |                  | GO_REF:0000033 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | plasma membrane                                                                                        |                      | GO_Central  | Homo sapiens | IBA      | PANTHER:PTN002798190<br>RGD:2059 <a href="#">more...</a>         | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein |                  | GO_REF:0000033 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | adenylate cyclase-activating adrenergic receptor signaling pathway                                     |                      | GO_Central  | Homo sapiens | IBA      | PANTHER:PTN001205212<br>UniProtKB:P08588 <a href="#">more...</a> | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein |                  | GO_REF:0000033 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | G protein-coupled dopamine receptor signaling pathway                                                  |                      | GO_Central  | Homo sapiens | IBA      | PANTHER:PTN008610765<br>RGD:2523<br>UniProtKB:P21918             | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein |                  | GO_REF:0000033 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | response to amphetamine                                                                                |                      | Ensembl     | Homo sapiens | IEA      | UniProtKB:Q8BLD9<br>ensembl:ENSMUSP00000039691                   | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein | UniProtKB:P21918 | GO_REF:0000107 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | regulation of systemic arterial blood pressure by vasopressin                                          |                      | Ensembl     | Homo sapiens | IEA      | UniProtKB:Q8BLD9<br>ensembl:ENSMUSP00000039691                   | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein | UniProtKB:P21918 | GO_REF:0000107 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | norepinephrine-epinephrine vasoconstriction involved in regulation of systemic arterial blood pressure |                      | Ensembl     | Homo sapiens | IEA      | UniProtKB:Q8BLD9<br>ensembl:ENSMUSP00000039691                   | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein | UniProtKB:P21918 | GO_REF:0000107 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | adenylate cyclase-activating G protein-coupled receptor signaling pathway                              |                      | Ensembl     | Homo sapiens | IEA      | UniProtKB:Q8BLD9<br>ensembl:ENSMUSP00000039691                   | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein | UniProtKB:P21918 | GO_REF:0000107 |
| <input type="checkbox"/> <b>DRD5</b>  | D(1B) dopamine receptor |           | mating behavior                                                                                        |                      | Ensembl     | Homo sapiens | IEA      | UniProtKB:Q8BLD9<br>ensembl:ENSMUSP00000039691                   | adrenergic receptor-related g-protein coupled receptor pthr24248 | protein | UniProtKB:P21918 | GO_REF:0000107 |

[https://amigo.geneontology.org/amigo/gene\\_product/UniProtKB:P21918](https://amigo.geneontology.org/amigo/gene_product/UniProtKB:P21918)

# STRING network structure



## Nodes:

Network nodes represent proteins

*splice isoforms or post-translational modifications are collapsed, i.e. each node represents all the proteins produced by a single, protein-coding gene locus.*

## Edges:

Edges represent protein-protein associations

*associations are meant to be specific and meaningful, i.e. proteins jointly contribute to a shared function; this does not necessarily mean they are physically binding each other.*

### Node Color



*colored nodes:  
query proteins and first shell of interactors*



*white nodes:  
second shell of interactors*

### Node Content



*empty nodes:  
proteins of unknown 3D structure*



*filled nodes:  
some 3D structure is known or predicted*

### Known Interactions



*from curated databases*



*experimentally determined*

### Predicted Interactions



*gene neighborhood*



*gene fusions*



*gene co-occurrence*

### Others



*textmining*



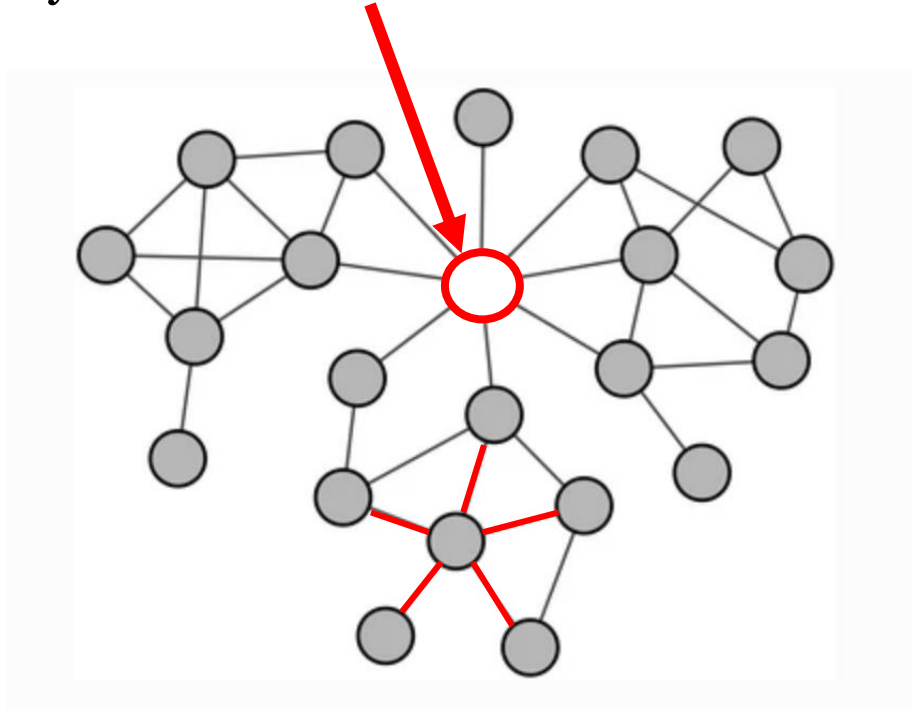
*co-expression*



*protein homology*

# Hubs – Betweenness centrality

Removal of a few **key hubs** disconnects the system into small isolated node clusters:

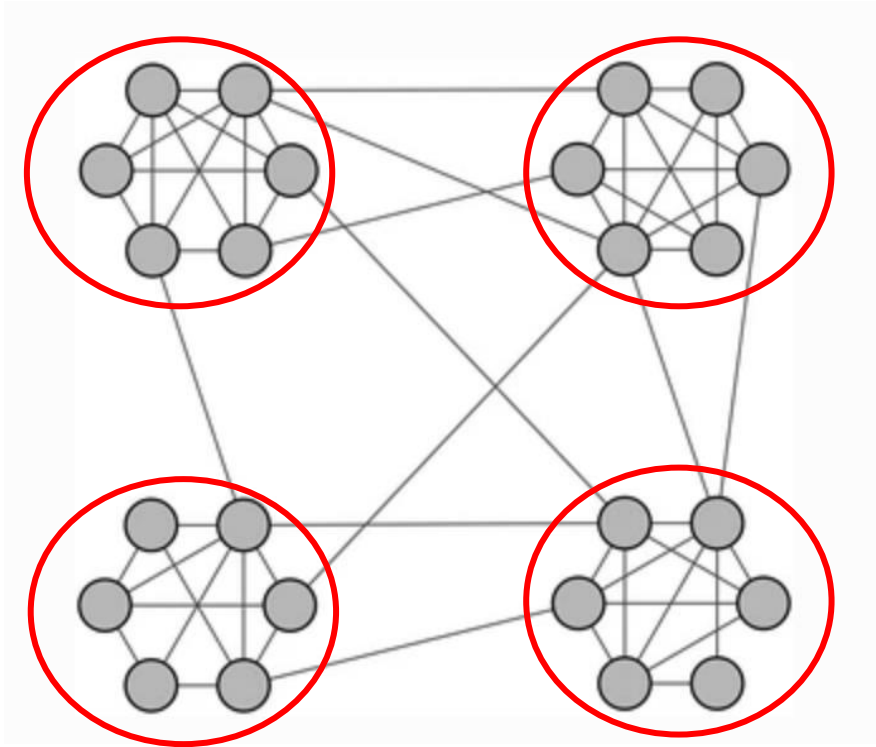


**Betweenness centrality:** role of a node in allowing information to pass from one part of the network to the other – shorter path

Degree: number of connections of a node ( 5 )

# Clusters

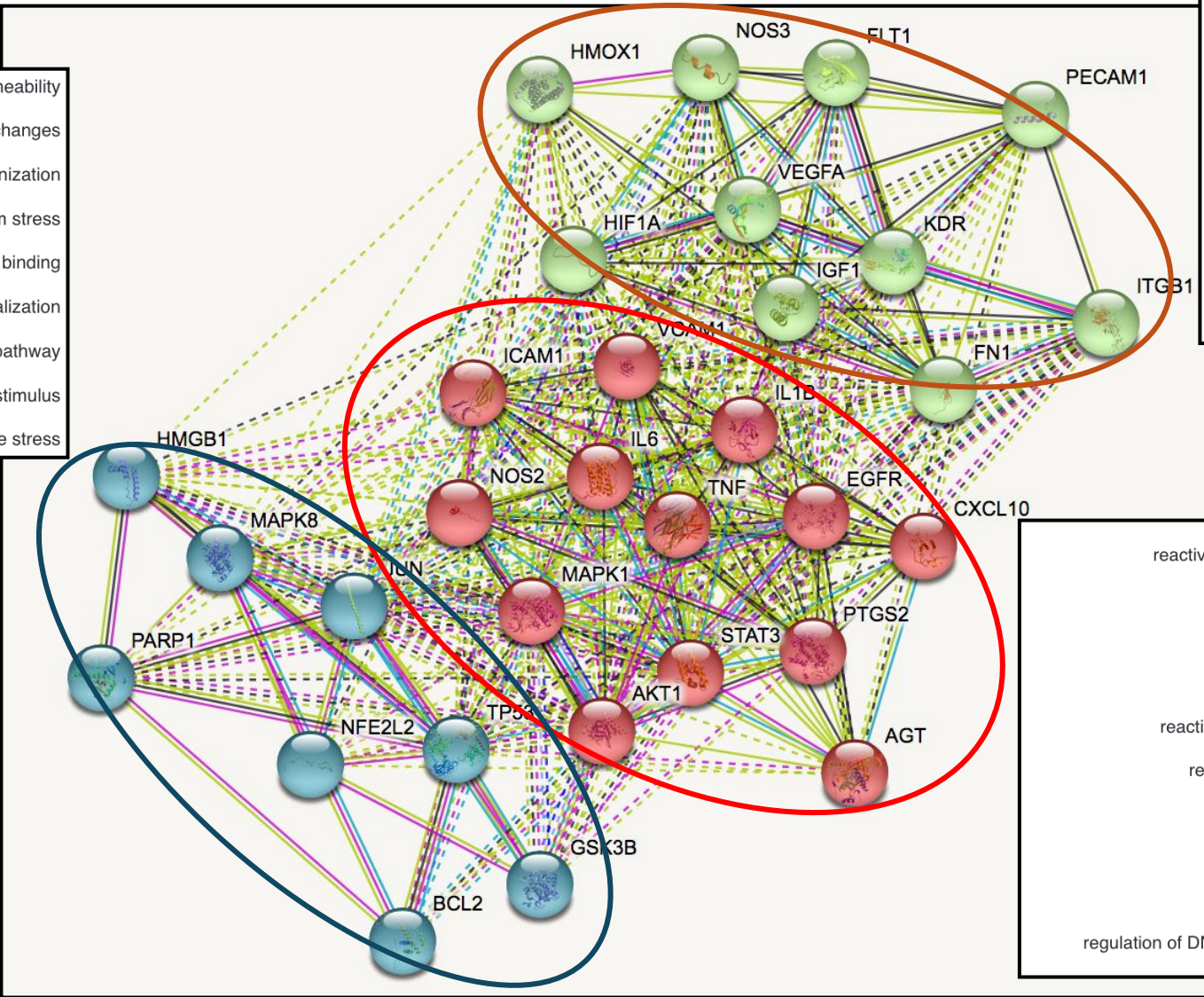
A subset in a graph with common characteristics or related to each other in particular manner



**Modularity:** A network with four distinct modules, each of which is highly connected but only lightly connected to other modules

# STRING MCL Clustering

- regulation of membrane permeability
- apoptotic mitochondrial changes
- mitochondrial membrane organization
- response to endoplasmic reticulum stress
- regulation of binding
- positive regulation of cellular protein localization
- regulation of apoptotic signaling pathway
- cellular response to external stimulus
- response to oxidative stress



- vascular endothelial growth factor receptor signaling pathway
- platelet degranulation
- tissue migration
- regulation of vasculature development
- ameboidal-type cell migration
- angiogenesis
- positive regulation of cell motility
- epithelial cell proliferation
- cell-substrate adhesion
- leukocyte migration

- reactive nitrogen species metabolic process
- acute inflammatory response
- muscle cell proliferation
- cellular response to biotic stimulus
- reactive oxygen species metabolic process
- response to molecule of bacterial origin
- gliogenesis
- cellular response to drug
- positive regulation of cell motility
- regulation of DNA-binding transcription factor activity

# The Markov Cluster Algorithm, aka the MCL algorithm

Inflation parameter effect

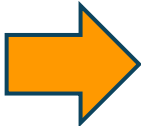


| Inflation Value | Granularity | Cluster Count | Cluster Size |
|-----------------|-------------|---------------|--------------|
| Low (1.2–1.4)   | Coarse      | Fewer         | Larger       |
| Default (~2.0)  | Balanced    | Moderate      | Medium       |
| High (4.0–6.0+) | Fine        | More          | Smaller      |

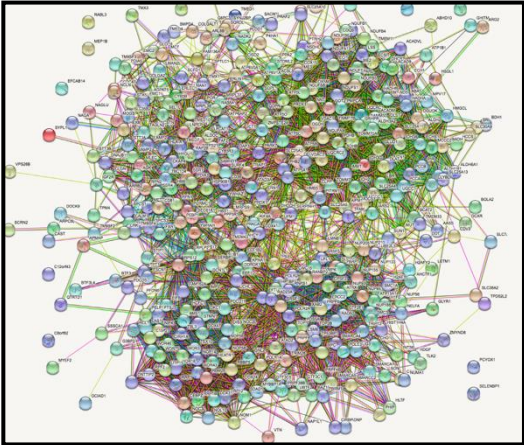
# Interaction network analysis

## User list

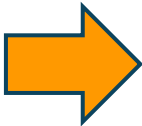
| A           |
|-------------|
| Gene symbol |
| HBB         |
| VTN         |
| ATP5L       |
| HBA1        |
| PBRM1       |
| MCCC1       |
| TIMM9       |
| TMED4       |
| GTF3C1      |
| EIF4EBP1    |
| STMN1       |
| HYPK        |
| TIMM13      |
| SF3B5       |
| VDAC3       |
| MCCC2       |
| PFDN4       |



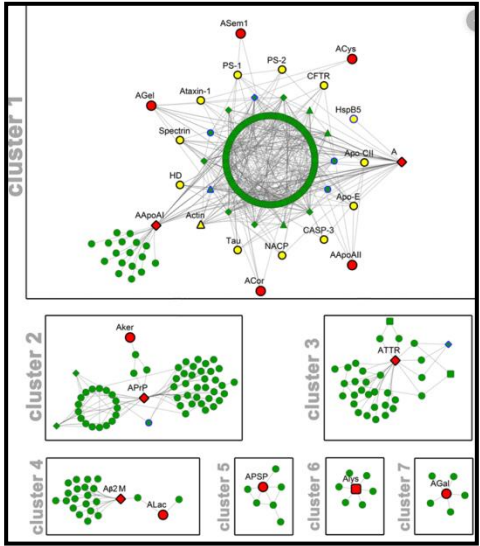
## PPI Database



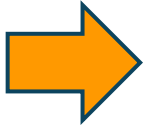
STRING  
BioGRID  
IntAct



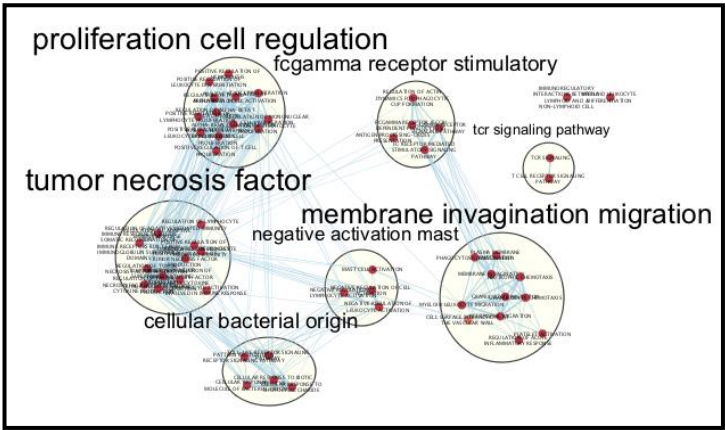
## Cluster analysis



STRING-MCL  
MCODE  
ClusterONE  
clusterMaker



## Cluster annotation



EnrichmentMap

# DEMO

## GSE125583

RNA from fusiform gyrus of Alzheimer's or neurologically normal post-mortem tissue.

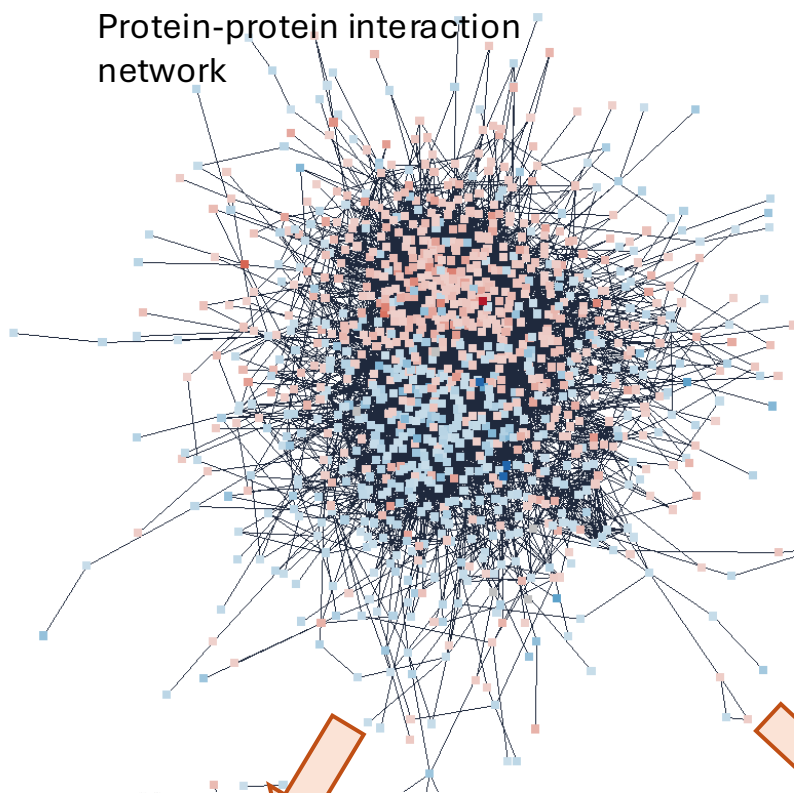
## DESeq2

- female AD vs female control
- Male AD vs male control  Abs Log2 foldchange > 0.5, padj<0.05  
(2163 DEGs)

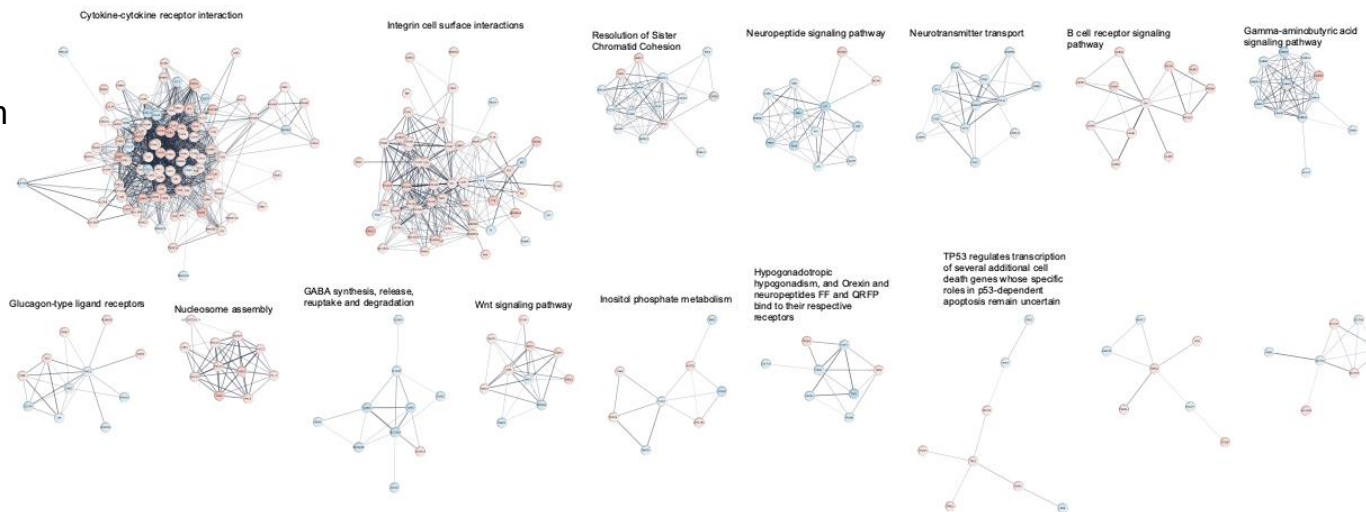
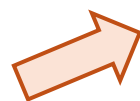
# Cytoscape stringApp

# DEMO

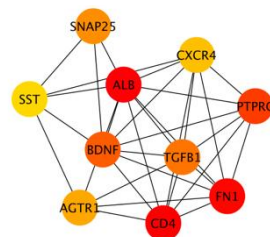
Protein-protein interaction network



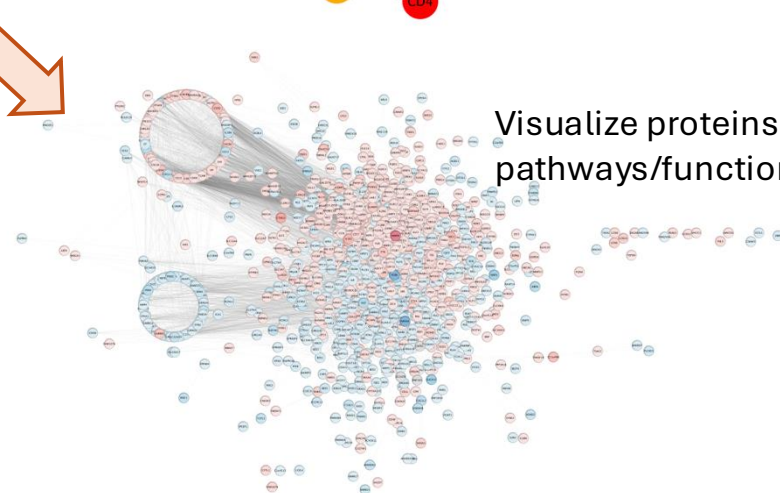
Clustering  
functional  
annotation



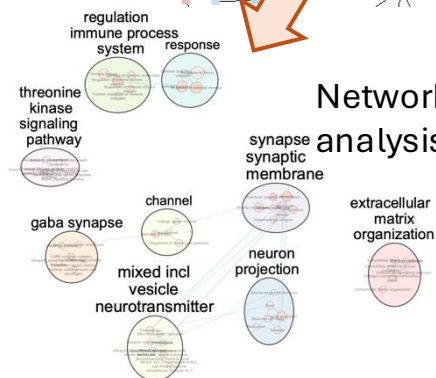
Identify hubs



Visualize proteins in  
pathways/functions



Network enrichment  
analysis



# STRING database Gene Set Enrichment Analysis

Protein by name >

Multiple proteins >

Proteins by sequences >

Proteins with Values/Ranks >

Protein families ("COGs") >

Pathway / Process / Disease >

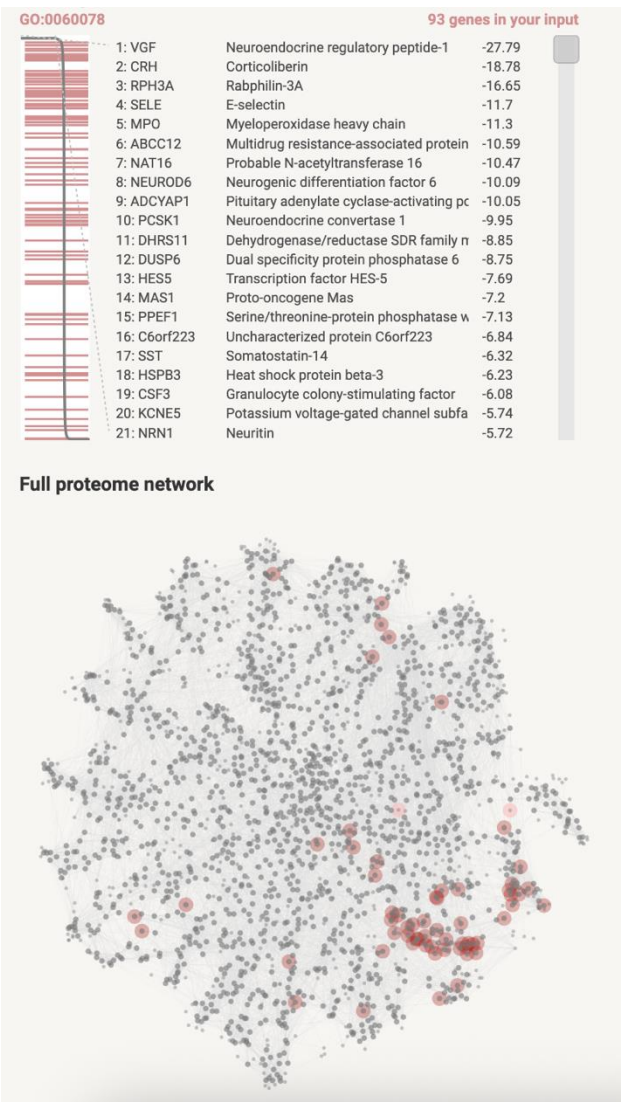
Add organism >

Organisms >

STRING chat New >

Examples >

Random entry >



| explain columns                    |                                                  |                    |                 |                      |                      |
|------------------------------------|--------------------------------------------------|--------------------|-----------------|----------------------|----------------------|
| Biological Process (Gene Ontology) |                                                  |                    |                 |                      |                      |
| GO-term                            | description                                      | * enrichment score | direction       | count in set/pathway | false discovery rate |
| GO:0060078                         | Regulation of postsynaptic membrane potential    | 0.00546757         | top of input    | 93 of 107            | 0.0067 (ks)          |
| GO:0032984                         | Protein-containing complex disassembly           | 0.00698905         | top of input    | 127 of 137           | 0.0040 (ks)          |
| GO:0140053                         | Mitochondrial gene expression                    | 0.0232891          | top of input    | 141 of 143           | 0.0034 (ks)          |
| GO:0051650                         | Establishment of vesicle localization            | 0.0246389          | top of input    | 165 of 169           | 0.0043 (ks)          |
| GO:0051648                         | Vesicle localization                             | 0.0276273          | top of input    | 176 of 180           | 0.00011 (ks)         |
| GO:0070972                         | Protein localization to endoplasmic reticulum    | 0.0280691          | top of input    | 66 of 66             | 0.0034 (ks)          |
| GO:0002181                         | Cytoplasmic translation                          | 0.0285379          | bottom of input | 122 of 123           | 6.44e-05 (ks)        |
| GO:0033108                         | Mitochondrial respiratory chain complex assembly | 0.0336414          | top of input    | 97 of 98             | 0.00027 (ks)         |
| GO:0032543                         | Mitochondrial translation                        | 0.0413337          | top of input    | 110 of 112           | 2.27e-05 (ks)        |
| GO:0008306                         | Associative learning                             | 0.0448071          | top of input    | 79 of 83             | 0.0027 (ks)          |
| GO:0009260                         | Ribonucleotide biosynthetic process              | 0.0450822          | top of input    | 200 of 211           | 0.0010 (ks)          |
| GO:0009260                         | Ribonucleotide biosynthetic process              | 0.0450822          | top of input    | 200 of 211           | 0.0010 (ks)          |

<https://version-12-0.string-db.org/cgi/globalenrichment?networkId=bZI5sov0bZ4N>

# Examples of data visualizations



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## Additional references and tutorials

- Dančík, V., Basu, A., Clemons, P. (2013). Properties of Biological Networks. In: Prokop, A., Csukás, B. (eds) Systems Biology. Springer, Dordrecht. [https://doi.org/10.1007/978-94-007-6803-1\\_5](https://doi.org/10.1007/978-94-007-6803-1_5)
- Chin CH, Chen SH, Wu HH, Ho CW, Ko MT, Lin CY. cytoHubba: identifying hub objects and sub-networks from complex interactome. BMC Syst Biol. 2014;8 Suppl 4(Suppl 4):S11. doi: 10.1186/1752-0509-8-S4-S11. Epub 2014 Dec 8. PMID: 25521941; PMCID: PMC4290687
- Isserlin R, Merico D, Voisin V, Bader GD. Enrichment Map - a Cytoscape app to visualize and explore OMICs pathway enrichment results. F1000Res. 2014 Jul 1;3:141. doi: 10.12688/f1000research.4536.1. PMID: 25075306; PMCID: PMC4103489.
- Kucera M, Isserlin R, Arkhangorodsky A, Bader GD. AutoAnnotate: A Cytoscape app for summarizing networks with semantic annotations. F1000Res. 2016 Jul 15;5:1717. doi: 10.12688/f1000research.9090.1. PMID: 27830058; PMCID: PMC5082607
- Shannon P, Markiel A, Ozier O, Baliga NS, Wang JT, Ramage D, Amin N, Schwikowski B, Ideker T. Cytoscape: a software environment for integrated models of biomolecular interaction networks. Genome Res. 2003 Nov;13(11):2498-504. doi: 10.1101/gr.1239303. PMID: 14597658; PMCID: PMC403769
- Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryary F, Hachilif R, Gable AL, Fang T, Doncheva NT, Pyysalo S, Bork P, Jensen LJ, von Mering C. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. Nucleic Acids Res. 2023 Jan 6;51(D1):D638-D646. doi: 10.1093/nar/gkac1000. PMID: 36370105; PMCID: PMC9825434

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Thank you!

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